# Giving Life to Private (Rated) Credit


Isabella Gschossmann
Imperial College

Ziang Li
Imperial College

Ali Uppal
Imperial College

Derek Wenning
Indiana University


**Wabash River Conference, August 2026**

- Increasing interest in **private assets** on institutional investors' balance sheets
  (FSOC 2025, SEC 2025, Senator Warren 2025, Treasury 2026)

- Why? Opaque **credit certification** $\rightarrow$ undermines risk-sensitive regulation

- Increasing interest in **private assets** on institutional investors' balance sheets (FSOC 2025, SEC 2025, Senator Warren 2025, Treasury 2026)
- Why? Opaque **credit certification** $\rightarrow$ undermines risk-sensitive regulation
- **Questions**
  1. Do private ratings offer a source of capital relief?
  2. How do private ratings affect RBC policy efficacy?
  3. Might demand for private ratings inadvertently create demand for private debt/credit?

- Increasing interest in **private assets** on institutional investors' balance sheets
  (FSOC 2025, SEC 2025, Senator Warren 2025, Treasury 2026)

- Why? Opaque **credit certification** $\rightarrow$ undermines risk-sensitive regulation

- **Questions**
  1. Do private ratings offer a source of capital relief?
  2. How do private ratings affect RBC policy efficacy?
  3. Might demand for private ratings inadvertently create demand for private debt/credit?

- **Setting:** the U.S. life insurance industry
  - largest institutional investor group in corporate bond markets
  - \$4 tril bond holdings — \$800 bil. *private* bond holdings — **\$500 bil. privately rated**

# Do Private Ratings Provide Capital Relief?

- Leverage two sources of variation using **bond-level** holdings data
  - Within-*issuer* $\rightarrow$ do priv. vs. pub. rated bonds of the same firm carry different ratings?
  - Within-*bond* $\rightarrow$ does switching out of private ratings $\uparrow$ capital charge?

# Do Private Ratings Provide Capital Relief?

- Leverage two sources of variation using **bond-level** holdings data
  - Within-*issuer* $\rightarrow$ do priv. vs. pub. rated bonds of the same firm carry different ratings?
  - Within-*bond* $\rightarrow$ does switching out of private ratings $\uparrow$ capital charge?

**Answer: Yes.**

Within-bond var'n $\rightarrow$ industry req. capital $\downarrow$ 6%, tail PLR holders $\downarrow\sim$ 30%

# Do Private Ratings Provide Capital Relief?

- Leverage two sources of variation using **bond-level** holdings data
  - Within-*issuer* $\rightarrow$ do priv. vs. pub. rated bonds of the same firm carry different ratings?
  - Within-*bond* $\rightarrow$ does switching out of private ratings $\uparrow$ capital charge?

**Answer: Yes.**

Within-bond var'n $\rightarrow$ industry req. capital $\downarrow$ 6%, tail PLR holders $\downarrow\sim$ 30%

Within-rating $\rightarrow$ yields 90bps $\uparrow$ rel. to public rating

## Do Priv Ratings (i) Mitigate Policy and (ii) Affect Private Debt Issuance?

- 2021 bond-based RBC reform $\rightarrow$ exogenous shock to capital requirements
  - 6 $\rightarrow$ 20 risk categories, broad $\rightarrow$ granular ratings-based schedule
  - Identification $\rightarrow$ pre-reform bond portfolio + randomness of within category charges

## Do Priv Ratings (i) Mitigate Policy and (ii) Affect Private Debt Issuance?

- 2021 bond-based RBC reform $\rightarrow$ exogenous shock to capital requirements
  - 6 $\rightarrow$ 20 risk categories, broad $\rightarrow$ granular ratings-based schedule
  - Identification $\rightarrow$ pre-reform bond portfolio + randomness of within category charges

**(i) Answer: Yes.**

Private ratings **attenuate RBC policies** through rebalancing (reach-for-opacity)

- 1SD shock to capital charges $\rightarrow$ $\sim$1pp (50% relative) tilt toward private ratings
- Ratings inflation $\rightarrow$ effect of policy on de-risking $\downarrow$ $\sim$50%

## Do Priv Ratings (i) Mitigate Policy and (ii) Affect Private Debt Issuance?

- 2021 bond-based RBC reform $\rightarrow$ exogenous shock to capital requirements
  - 6 $\rightarrow$ 20 risk categories, broad $\rightarrow$ granular ratings-based schedule
  - Identification $\rightarrow$ pre-reform bond portfolio + randomness of within category charges

**(i) Answer: Yes.**

Private ratings **attenuate RBC policies** through rebalancing (reach-for-opacity)

- 1SD shock to capital charges $\rightarrow$ $\sim$1pp (50% relative) tilt toward private ratings
- Ratings inflation $\rightarrow$ effect of policy on de-risking $\downarrow$ $\sim$50%

**(ii) Answer: Yes.**

Private rating **demand** $\rightarrow$ private bond **issuance**

- Firms 1SD higher connection to exposed insurers $\uparrow$ private issuance $\sim$16%

# Literature

- **Credit Ratings in Opaque Markets**

  Post-GFC: Skreta & Veldkamp 2009, Benmelech & Dlugosz 2010, Bolton et al 2012, Griffin & Tang 2012

  Ratings Agencies: Opp et al 2013, Dimitrov et al 2015, Bruno et al 2016, Behr et al 2018

  Regulation: Acharya et al 2013, Mariathasan & Merrouche 2014, Plosser & Santos 2018, Behn et al 2022

  **This paper:** *External* ratings & *investor-driven* decisions, even without internal modeling

- **Insurers, Regulation, Portfolio Choice, and Private Markets**

  Private Markets: Kirti & Sarin 2024, Meisenzahl et al 2025, Carlino et al 2025, Huber et al 2026

  Portfolio Choice: Ellul et al 2011, Ge & Weisbach 2021, Becker et al 2022, Bhardwaj et al 2025, Li 2026

  Valuation: Li, Oh, & Ricciardi 2026, Sen, Sharma, & Viceira 2026

  Regulatory Arbitrage: Becker & Ivashina 2015, Merrill et al 2019, Fringuellotti & Santos 2021, Weinlich 2024

  **This paper:** Full pipeline between capital charge → rebalancing → spillovers, focus on private markets

# Institutional Details

# Risk-Based Capital Timeline

- RBC framework was introduced in 1993
  - Featured **broad** risk designations for bonds mapped to **credit ratings**
    Example: AAA through A- = NAIC 1 (39 bps charge)

# Risk-Based Capital Timeline

- RBC framework was introduced in 1993
  - Featured <u>**broad**</u> risk designations for bonds mapped to **credit ratings**
    Example: AAA through A- = NAIC 1 (39 bps charge)

- Regulators realized desig's were <u>*too*</u> broad, hiding risk (Becker & Ivashina, 2015)

  pre-2020: broad discussion about *granularizing* the existing system, no real details

## Risk-Based Capital Timeline

- RBC framework was introduced in 1993
  - Featured **broad** risk designations for bonds mapped to **credit ratings**
    Example: AAA through A- = NAIC 1 (39 bps charge)

- Regulators realized desig's were *too* broad, hiding risk (Becker & Ivashina, 2015)

  pre-2020: broad discussion about *granularizing* the existing system, no real details

  |

  2020: ACLI commissions Moody's to recommend new factors

# Risk-Based Capital Timeline

- RBC framework was introduced in 1993
  - Featured **broad** risk designations for bonds mapped to **credit ratings**
    Example: AAA through A- = NAIC 1 (39 bps charge)

- Regulators realized desig's were *too* broad, hiding risk (Becker & Ivashina, 2015)

pre-2020: broad discussion about *granularizing* the existing system, no real details

|

2020: ACLI commissions Moody's to recommend new factors

|

2021 (Feb): Moody's delivers report

## Risk-Based Capital Timeline

- RBC framework was introduced in 1993
  - Featured **broad** risk designations for bonds mapped to **credit ratings**
    Example: AAA through A- = NAIC 1 (39 bps charge)

- Regulators realized desig's were *too* broad, hiding risk (Becker & Ivashina, 2015)

  pre-2020: broad discussion about *granularizing* the existing system, no real details
  |
  2020: ACLI commissions Moody's to recommend new factors
  |
  2021 (Feb): Moody's delivers report
  |
  2021 (Apr): NAIC proposes new charges (AAA+ $\rightarrow$ 16 bps, A- $\rightarrow$ 102 bps, etc)

## Risk-Based Capital Timeline

- RBC framework was introduced in 1993
  - Featured **broad** risk designations for bonds mapped to **credit ratings**
    Example: AAA through A- = NAIC 1 (39 bps charge)

- Regulators realized desig's were *too* broad, hiding risk (Becker & Ivashina, 2015)

pre-2020: broad discussion about *granularizing* the existing system, no real details
|
2020: ACLI commissions Moody's to recommend new factors
|
2021 (Feb): Moody's delivers report
|
2021 (Apr): NAIC proposes new charges (AAA+ $\rightarrow$ 16 bps, A- $\rightarrow$ 102 bps, etc)
|
2021 (Jun): NAIC adopts proposal, implements with 2021Q4 filings

# How are Risk Designations Determined?

- Insurers file their bond holdings with the *Securities Valuation Office (SVO)*

- For **publicly rated** bonds: if rating is reputable, bond is "filing exempt"
  $\rightarrow$ Direct mapping from credit rating to risk factor

## How are Risk Designations Determined?

- Insurers file their bond holdings with the *Securities Valuation Office (SVO)*

- For **publicly rated** bonds: if rating is reputable, bond is "filing exempt"
  → Direct mapping from credit rating to risk factor

- **Private ratings** are different — require (since 2022) a separate report
  - Often affiliated with private or opaque assets
  - Typically written by small, niche credit rating agencies
  - *NAIC Working Group E — functionality for review was limited until YE 2024*

## How are Risk Designations Determined?

- Insurers file their bond holdings with the *Securities Valuation Office (SVO)*
- For **publicly rated** bonds: if rating is reputable, bond is "filing exempt"
  $\rightarrow$ Direct mapping from credit rating to risk factor
- **Private ratings** are different — require (since 2022) a separate report
  - Often affiliated with private or opaque assets
  - Typically written by small, niche credit rating agencies
  - *NAIC Working Group E — functionality for review was limited until YE 2024*

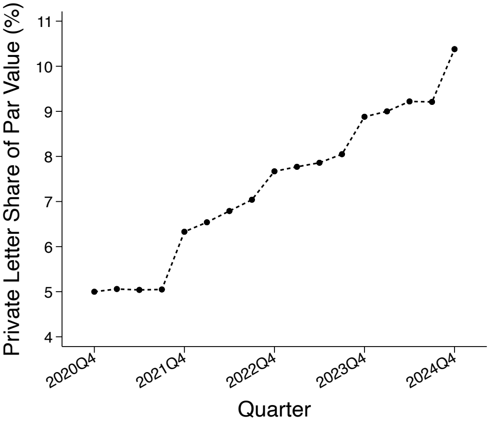
**Assessement:** great for ratings arb!

## Data

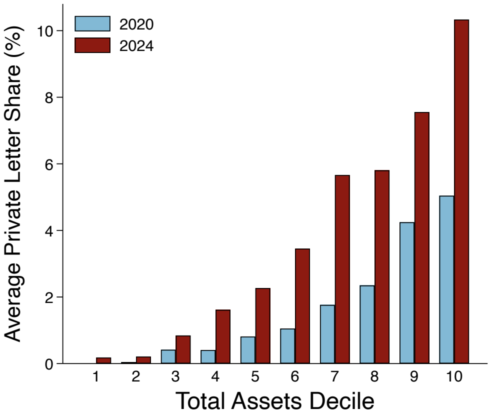
- **NAIC Schedule D** — insurer $\times$ bond holdings panel, 2020-2024
  - Key vars: par value (qtr)/CV (yr), yields, risk factors, SVO Symbol (PL)
  - Extras: callability, cap. structure, maturity, PP dummy, asset type
- **Mergent FISD & Refinitiv** — firm-level bond issuance panel, 2018-2024
  - "PP" dummy — typically "no SEC filing" or 144A (not exact def used by insurers)
- **WRDS** — public bond market yield panel
  - used for robustness only

# Characteristics of Bonds with Private Letters

# Fact 1a: Private Letter Holdings Doubled in 5 Years

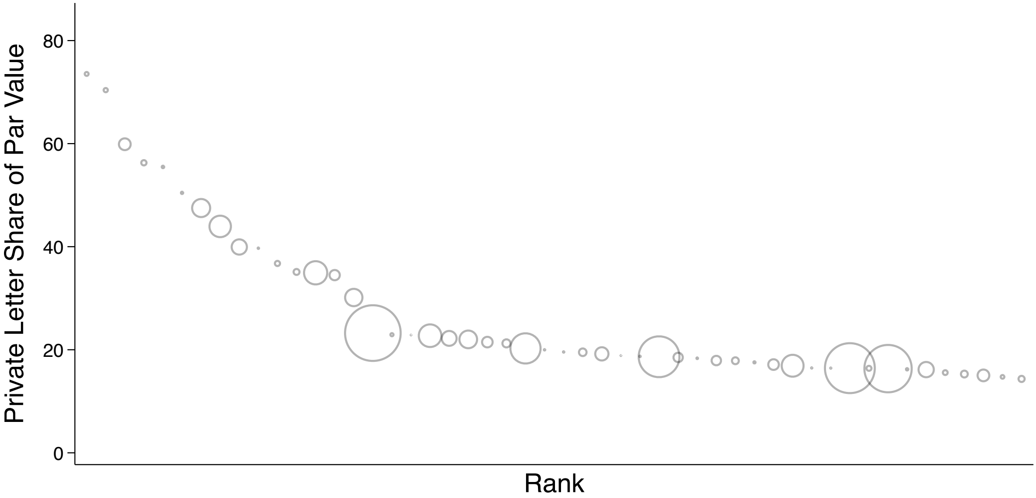



(a) Aggregate Growth in PL Holdings

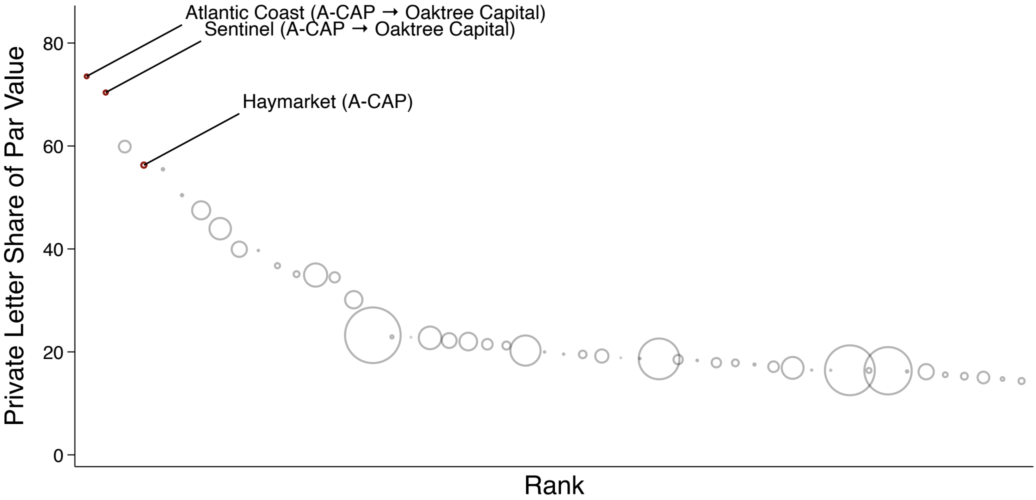



(b) Growth by Size Decile

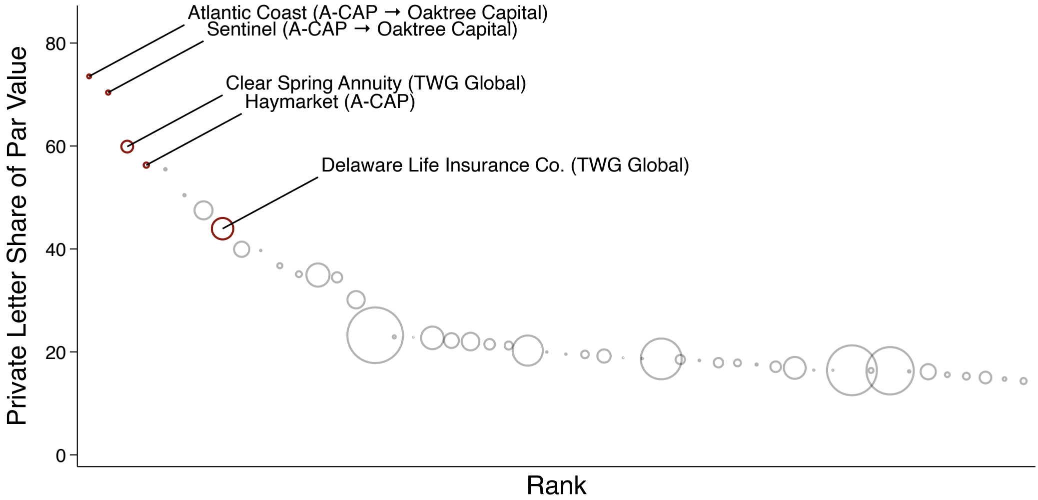
# Fact 1b: Private Letter Holdings Have a Long Right Tail



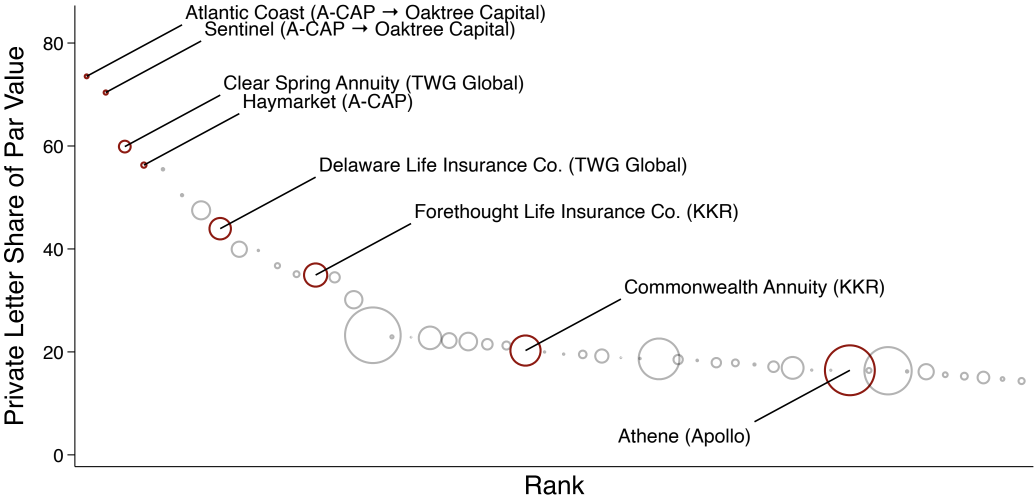
# Fact 1b: Private Letter Holdings Have a Long Right Tail



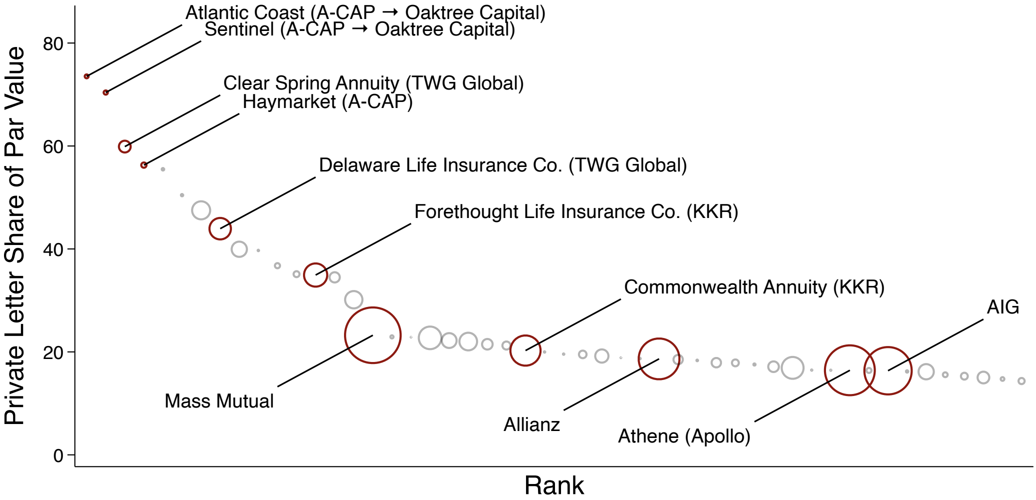
# Fact 1b: Private Letter Holdings Have a Long Right Tail



# Fact 1b: Private Letter Holdings Have a Long Right Tail



# Fact 1b: Private Letter Holdings Have a Long Right Tail



# Fact 2: Bonds with Private Ratings are Different

| | Rating Type | | |
|---|---|---|---|
| *% of par value that is...* | Private | Public | Difference |

# Fact 2: Bonds with Private Ratings are Different

| | Rating Type | | |
|---|---|---|---|
| *% of par value that is...* | Private | Public | Difference |
| Privately Placed | 81.5 | 14.1 | **67.4** |

## Fact 2: Bonds with Private Ratings are Different

| *% of par value that is...* | Rating Type: Private | Rating Type: Public | Difference |
|---|---|---|---|
| Privately Placed | 81.5 | 14.1 | **67.4** |
| Senior Secured | 42.0 | 25.3 | **16.7** |
| Variable Rate | 25.3 | 13.6 | **11.7** |
| Callable | 20.0 | 69.8 | **-49.8** |

# Fact 2: Bonds with Private Ratings are Different

| | Rating Type | | |
|---|---|---|---|
| *% of par value that is...* | Private | Public | Difference |
| Privately Placed | 81.5 | 14.1 | **67.4** |
| Senior Secured | 42.0 | 25.3 | **16.7** |
| Variable Rate | 25.3 | 13.6 | **11.7** |
| Callable | 20.0 | 69.8 | **-49.8** |
| Maturity $< 5$ | 23.2 | 18.5 | **4.7** |
| Maturity $\in [5, 10)$ | 28.2 | 22.5 | **5.7** |

# Fact 2: Bonds with Private Ratings are Different

| *% of par value that is...* | Rating Type: Private | Rating Type: Public | Difference |
|---|---|---|---|
| Privately Placed | 81.5 | 14.1 | **67.4** |
| Senior Secured | 42.0 | 25.3 | **16.7** |
| Variable Rate | 25.3 | 13.6 | **11.7** |
| Callable | 20.0 | 69.8 | **-49.8** |
| Maturity $< 5$ | 23.2 | 18.5 | **4.7** |
| Maturity $\in [5, 10)$ | 28.2 | 22.5 | **5.7** |
| Issuer Obligation | 61.1 | 74.3 | **-13.2** |
| Loan-Backed/Structured Security | 24.9 | 13.1 | **11.8** |

## Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$\text{(RBC Risk Factor)}_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | |
|---|---|
| Private Rating | |
| Observations | |
| Within $R^2$ | |

+Time-varying controls   +Insurer Disagreements

## Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$(\text{RBC Risk Factor})_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | |
|---|---|
| Private Rating | 0.85*** |
| | (0.13) |
| Year FE | ✓ |
| Maturity FE | ✓ |
| Observations | 236,358 |
| Within $R^2$ | 0.001 |

# Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$(\text{RBC Risk Factor})_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | | |
|---|---|---|
| Private Rating | 0.85*** | −1.94*** |
| | (0.13) | (0.32) |
| Year FE | ✓ | ✓ |
| Issuer FE | | ✓ |
| Maturity FE | ✓ | ✓ |
| Observations | 236,358 | 232,718 |
| Within $R^2$ | 0.001 | 0.002 |

# Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$(\text{RBC Risk Factor})_{ibt} = \beta\ (\textbf{Private Rating})_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | | | |
|---|---|---|---|
| Private Rating | $0.85^{***}$ | $-1.94^{***}$ | $-1.92^{***}$ |
| | (0.13) | (0.32) | (0.44) |
| Year FE | ✓ | ✓ | |
| Issuer FE | | ✓ | |
| Issuer × Year FE | | | ✓ |
| Maturity FE | ✓ | ✓ | ✓ |
| Observations | 236,358 | 232,718 | 195,741 |
| Within $R^2$ | 0.001 | 0.002 | 0.001 |

▸ +Time-varying controls ▸ +Insurer Disagreements

## Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$\text{(RBC Risk Factor)}_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | | | | |
|---|---|---|---|---|
| Private Rating | $0.85^{***}$ | $-1.94^{***}$ | $-1.92^{***}$ | $-1.94^{***}$ |
| | (0.13) | (0.32) | (0.44) | (0.46) |
| Year FE | ✓ | ✓ | | |
| Issuer FE | | ✓ | | |
| Issuer × Year FE | | | ✓ | ✓ |
| Bond Controls | | | | ✓ |
| Maturity FE | ✓ | ✓ | ✓ | ✓ |
| Observations | 236,358 | 232,718 | 195,741 | 176,656 |
| Within $R^2$ | 0.001 | 0.002 | 0.001 | 0.009 |

▸ +Time-varying controls ▸ +Insurer Disagreements

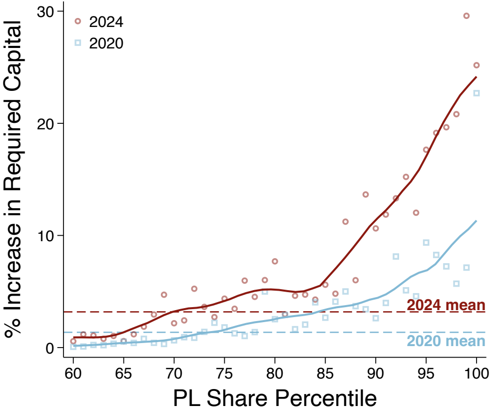
# Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$(\text{RBC Risk Factor})_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

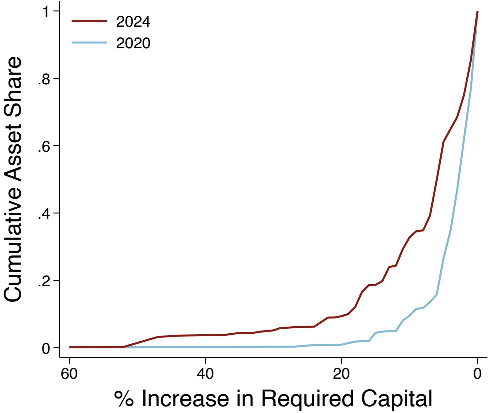
| | | | | | |
|---|---|---|---|---|---|
| Private Rating | $0.85^{***}$ | $-1.94^{***}$ | $-1.92^{***}$ | $-1.94^{***}$ | $-1.76^{***}$ |
| | (0.13) | (0.32) | (0.44) | (0.46) | (0.37) |
| Year FE | ✓ | ✓ | | | ✓ |
| Issuer FE | | ✓ | | | |
| Issuer × Year FE | | | ✓ | ✓ | |
| Bond Controls | | | | ✓ | |
| Bond FE | | | | | ✓ |
| Maturity FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 236,358 | 232,718 | 195,741 | 176,656 | 213,187 |
| Within $R^2$ | 0.001 | 0.002 | 0.001 | 0.009 | 0.005 |

▸ +Time-varying controls ▸ +Insurer Disagreements

# Fact 3b: Downgrading PLRs Would Increase Req. Capital Significantly




(a) Tail




(b) Cumulative Asset CDF

## Fact 4: Bonds with Private Ratings have Capital Efficient Returns

$$\text{Yield}_{ibt} = \beta \ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | |
|---|---|
| Private Rating | |
| | |
| Rating $\times$ Year FE | |
| Maturity FE | |
| | |
| Observations | |
| Within $R^2$ | |

## Fact 4: Bonds with Private Ratings have Capital Efficient Returns

$$\text{Yield}_{ibt} = \beta \ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | | |
|---|---|---|
| Private Rating | 1.44*** | 0.95*** |
| | (0.12) | (0.15) |
| Rating × Year FE | ✓ | ✓ |
| Maturity FE | ✓ | ✓ |
| Bond Controls | | ✓ |
| Insurer Filings | ✓ | ✓ |
| Observations | 62,791 | 29,355 |
| Within $R^2$ | 0.002 | 0.002 |

## Fact 4: Bonds with Private Ratings have Capital Efficient Returns

$$\text{Yield}_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | | | | |
|---|---|---|---|---|
| Private Rating | 1.44*** | 0.95*** | 2.95*** | 2.71*** |
| | (0.12) | (0.15) | (0.21) | (0.28) |
| Rating × Year FE | ✓ | ✓ | ✓ | ✓ |
| Maturity FE | ✓ | ✓ | ✓ | ✓ |
| Bond Controls | | ✓ | | ✓ |
| Insurer Filings | ✓ | ✓ | | |
| WRDS (public bonds) | | | ✓ | ✓ |
| Observations | 62,791 | 29,355 | 43,549 | 29,742 |
| Within $R^2$ | 0.002 | 0.002 | 0.004 | 0.005 |

## Summary of the Facts: Bonds with private ratings...

**(Fact 1)** ... have doubled their portfolio share on insurers' balance sheets in 5 years

**(Fact 2)** ... are often private in nature and are distinct from traditional bonds

**(Fact 3)** ... appear inflated, and the magnitude is meaningful for RBC

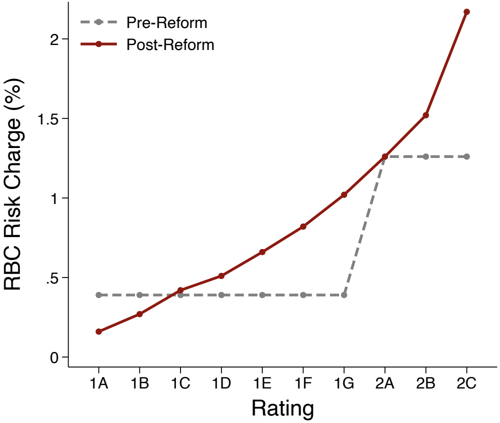
**(Fact 4)** ... provide capital-efficient yields relative to similarly-publicly-rated bonds

# Do insurers use private ratings for capital relief?

# The RBC Reform – an Exogenous Shock to Required Capital

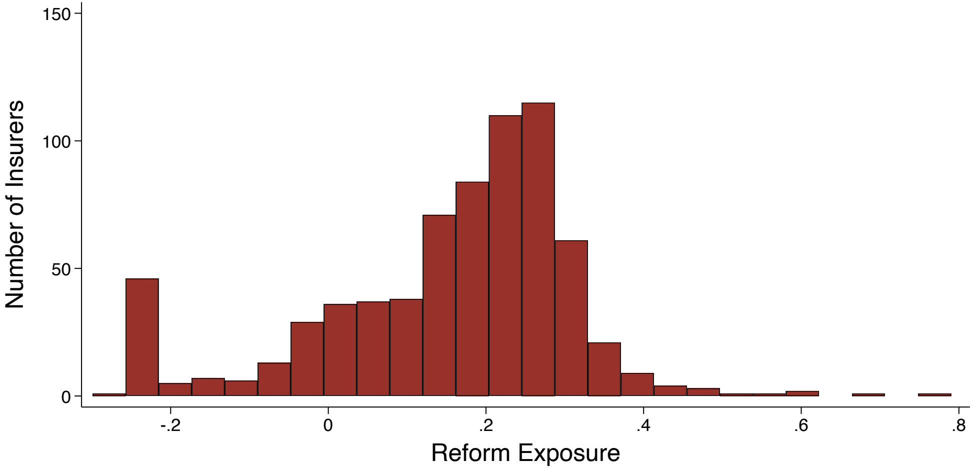
- Leverage exogenous *level* changes in risk-charges *across ratings*

- **Assumption:**
  **Pre-reform p'lio weights** *exogenous* to
  **Δ in risk-factor schedule**
  - Timing: reform announced mid-year
  - Levels: var'n in broad rating *buckets*



## Reform Exposure

$$\textbf{Exposure}_j = \sum_c \textbf{(bond portfolio weight)}_{jc,2020} \times \Delta\textbf{(risk-factor)}_c$$



# Did the RBC Reform Incentivize Private Letter Holdings?

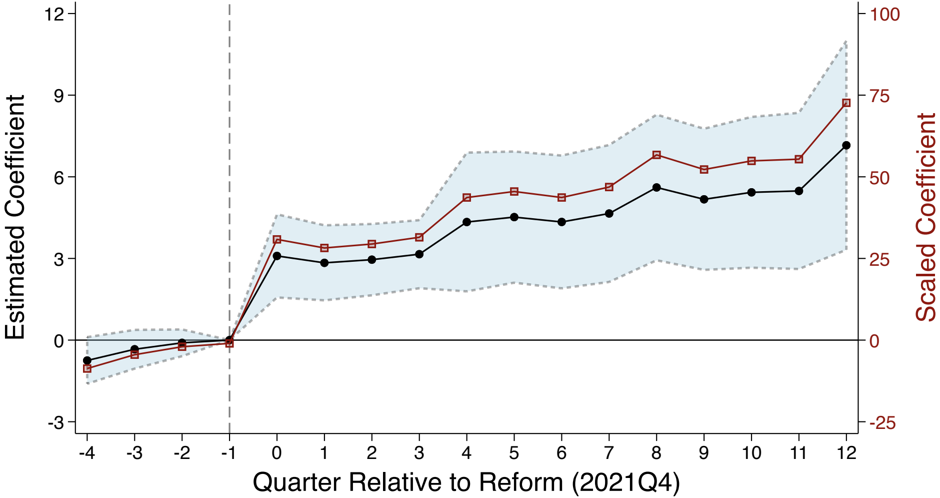
$$\text{SharePL}_{jt} = \beta \ \text{Exposure}_j \times \text{Post}_t + \text{controls} + \alpha_t + \delta_j + \varepsilon_{jt}$$

# Did the RBC Reform Incentivize Private Letter Holdings?

$$\text{SharePL}_{jt} = \beta \text{ Exposure}_j \times \text{Post}_t + \text{controls} + \alpha_t + \delta_j + \varepsilon_{jt}$$

| | | | | |
|---|---|---|---|---|
| Post | 0.62*** | | | |
| | (0.14) | | | |
| Exposure | 6.30*** | 6.29*** | | |
| | (1.08) | (1.08) | | |
| **Exposure × Post** | 4.89*** | 4.88*** | 4.66*** | 4.76*** |
| | (1.10) | (1.10) | (1.07) | (1.07) |
| Quarter FE | | ✓ | ✓ | ✓ |
| Insurer FE | | | ✓ | ✓ |
| Size + ROA Controls | | | | ✓ |
| Observations | 11,428 | 11,428 | 11,424 | 11,356 |
| Within $R^2$ | 0.08 | 0.07 | 0.02 | 0.02 |

# Did the RBC Reform Incentivize Private Letter Holdings?



## Did Private Ratings Attenuate the Effect of the Reform?

- **Exposure:** *given prior portfolio choice, how does reform affect total capital charge?*

# Did Private Ratings Attenuate the Effect of the Reform?

- **Exposure:** *given prior portfolio choice, how does reform affect total capital charge?*

- Regulators should expect **broad rebalancing** in response to large exposure
  - 1G (A-) bonds less capital efficient $\rightarrow$ 1G $\downarrow$, 1A $\uparrow$
  - Measure ex-*post* capital charge:

$$(\text{Ex-Post Charge})_{jt} = \sum_{b \in \text{bonds}} \boldsymbol{\omega_{jbt}} \times (\text{New RBC Factor})_b$$

# Did Private Ratings Attenuate the Effect of the Reform?

- **Exposure:** *given prior portfolio choice, how does reform affect total capital charge?*
- Regulators should expect **broad rebalancing** in response to large exposure
  - 1G (A-) bonds less capital efficient $\rightarrow$ 1G $\downarrow$, 1A $\uparrow$
  - Measure ex-*post* capital charge:

$$(\text{Ex-Post Charge})_{jt} = \sum_{b \in \text{bonds}} \omega_{jbt} \times (\text{New RBC Factor})_b$$

**Question:** *how much does shift into PLRs attenuate reported risk rebalancing?*

# Did Private Ratings Attenuate the Effect of the Reform?

- **Exposure:** *given prior portfolio choice, how does reform affect total capital charge?*

- Regulators should expect **broad rebalancing** in response to large exposure
  - 1G (A-) bonds less capital efficient $\rightarrow$ 1G $\downarrow$, 1A $\uparrow$
  - Measure ex-*post* capital charge:

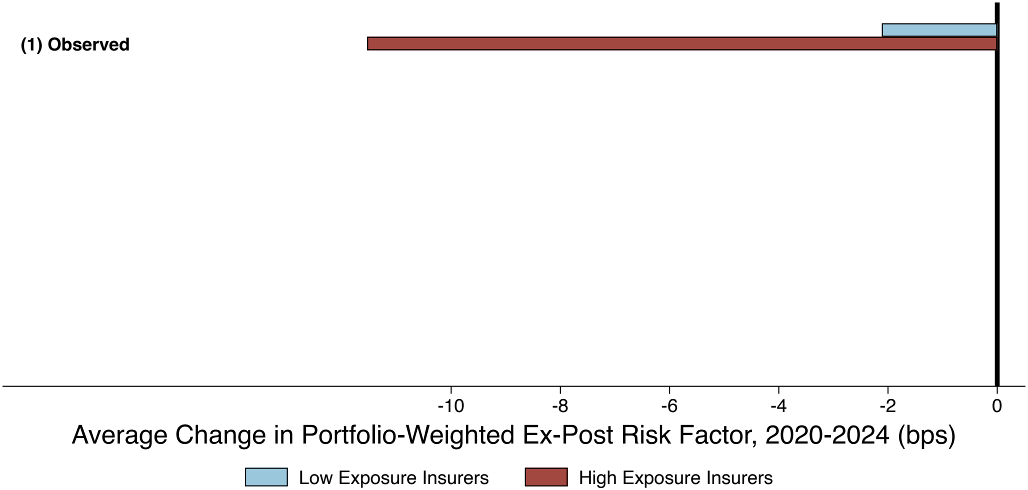
$$(\text{Ex-Post Charge})_{jt} = \sum_{b \in \text{bonds}} \omega_{jbt} \times (\text{New RBC Factor})_b$$

**Question:** *how much does shift into PLRs attenuate reported risk rebalancing?*

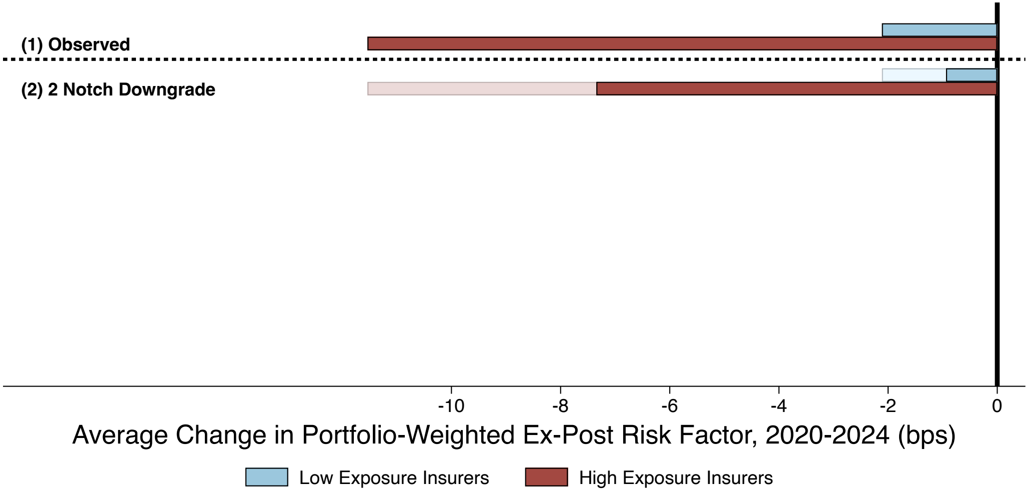
**Idea:** Use PL inflation estimates from Fact 3 to **downgrade** bonds

$\rightarrow$ 7 counterfactual charges using different specifications

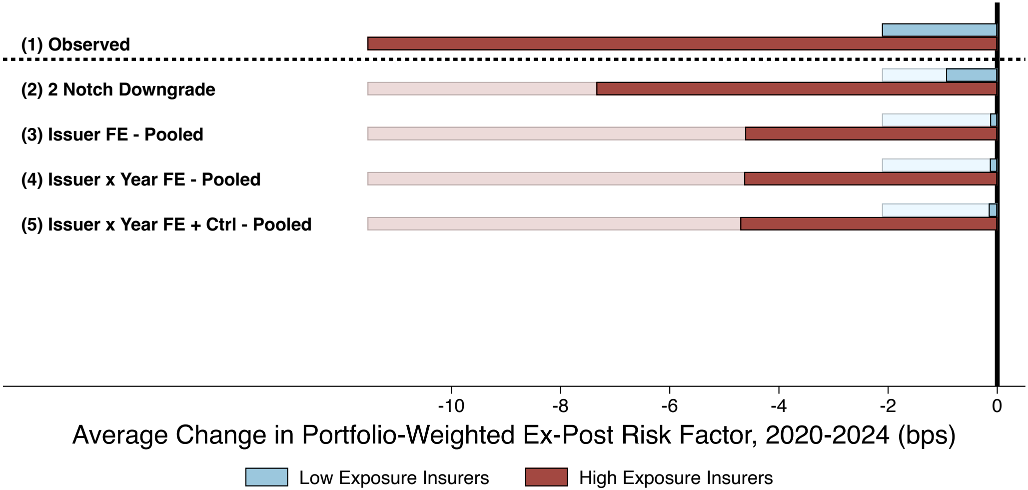
# Did Private Ratings Attenuate the Effect of the Reform?



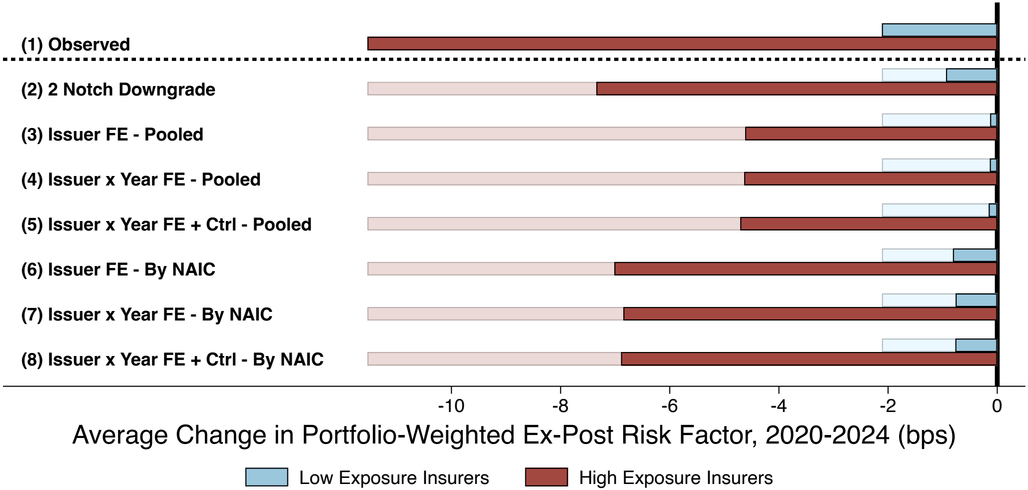
# Did Private Ratings Attenuate the Effect of the Reform?



# Did Private Ratings Attenuate the Effect of the Reform?

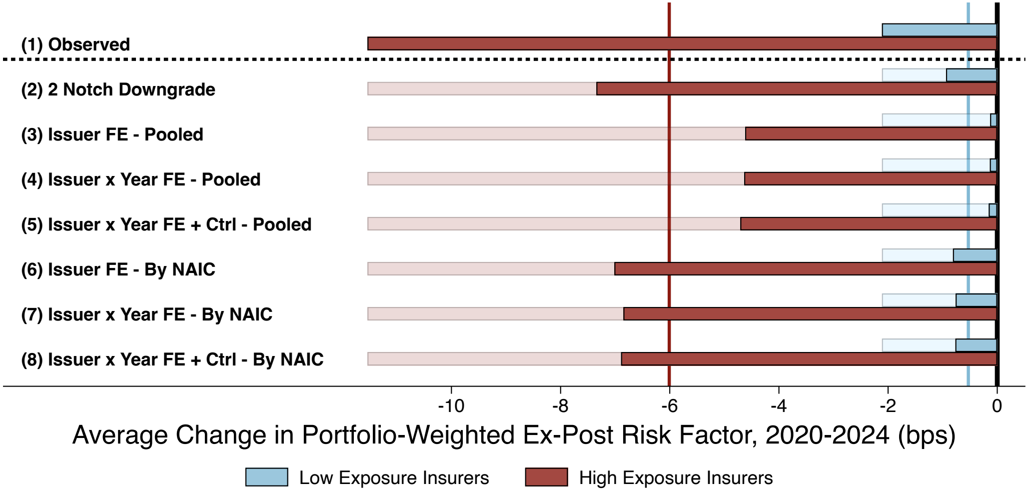


# Did Private Ratings Attenuate the Effect of the Reform?




Regressions

# Did Private Ratings Attenuate the Effect of the Reform?




Regressions

# Does Private Ratings Demand $\rightarrow$ Private Debt Issuance?

# Did the Reform Create Demand for Ratings or Placement?

$$(\text{Portfolio Share})_{jt} = \beta \text{ Exposed}_j \times \text{Post}_t + \alpha_t + \delta_j + \varepsilon_{jt}$$

| | |
|---|---|
| Exposure × Post | |
| | |
| Private Rating? | |
| Private Placement? | |
| Quarter FE | |
| Insurer FE | |
| Size + ROA Controls | |
| Observations | |
| Within $R^2$ | |

# Did the Reform Create Demand for Ratings or Placement?

$$(\text{Portfolio Share})_{jt} = \beta \text{ Exposed}_j \times \text{Post}_t + \alpha_t + \delta_j + \varepsilon_{jt}$$

| | |
|---|---|
| Exposure × Post | 2.19*** |
| | (0.78) |
| Private Rating? | ✓ |
| Private Placement? | |
| Quarter FE | ✓ |
| Insurer FE | ✓ |
| Size + ROA Controls | ✓ |
| Observations | 11,428 |
| Within $R^2$ | 0.08 |

# Did the Reform Create Demand for Ratings or Placement?

$$(\text{Portfolio Share})_{jt} = \beta\ \text{Exposed}_j \times \text{Post}_t + \alpha_t + \delta_j + \varepsilon_{jt}$$

| | | |
|---|---|---|
| Exposure × Post | 2.19*** | 2.58*** |
| | (0.78) | (0.71) |
| Private Rating? | ✓ | ✓ |
| Private Placement? | | ✓ |
| Quarter FE | ✓ | ✓ |
| Insurer FE | ✓ | ✓ |
| Size + ROA Controls | ✓ | ✓ |
| Observations | 11,428 | 11,428 |
| Within $R^2$ | 0.08 | 0.07 |

# Did the Reform Create Demand for Ratings or Placement?

$$(\text{Portfolio Share})_{jt} = \beta\ \text{Exposed}_j \times \text{Post}_t + \alpha_t + \delta_j + \varepsilon_{jt}$$

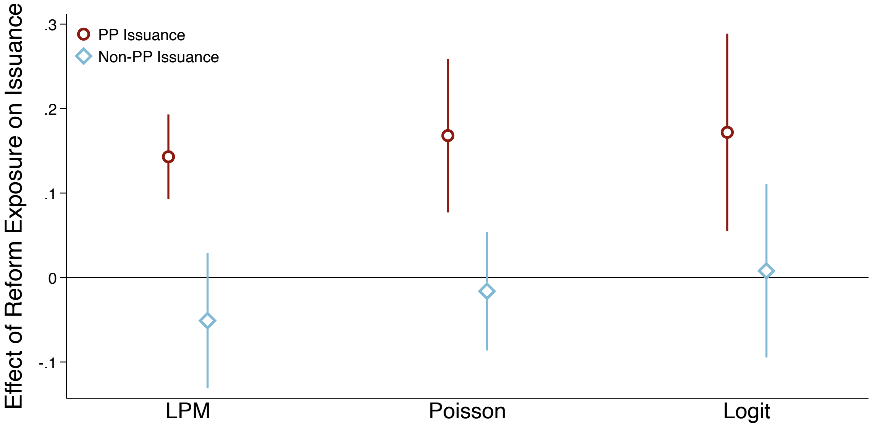
| | | | |
|---|---|---|---|
| Exposure × Post | 2.19*** | 2.58*** | 1.49 |
| | (0.78) | (0.71) | (0.97) |
| Private Rating? | ✓ | ✓ | |
| Private Placement? | | ✓ | ✓ |
| Quarter FE | ✓ | ✓ | ✓ |
| Insurer FE | ✓ | ✓ | ✓ |
| Size + ROA Controls | ✓ | ✓ | ✓ |
| Observations | 11,428 | 11,428 | 11,424 |
| Within $R^2$ | 0.08 | 0.07 | 0.02 |

## Did Private Rating Demand Increase Private Debt Issuance?

(Firm $i$'s **(non)** **PP** Issuance$)_t = \beta$(RBC Reform Exposure$)_i \times$ Post$_t + \alpha_i + \delta_t + \varepsilon_{it}$

## Did Private Rating Demand Increase Private Debt Issuance?

$$\text{(Firm } i\text{'s } \textbf{(non) PP} \text{ Issuance})_t = \beta(\text{RBC Reform Exposure})_i \times \text{Post}_t + \alpha_i + \delta_t + \varepsilon_{it}$$




‣ New PP → PL

# Conclusion

# Private Ratings Reduce Policy Efficacy and Expand Opaque Debt Issuance

- Insurers exploit ratings inflation in opaque bonds to sidestep capital requirements
  - Private ratings $\rightarrow$ capital efficient due to ratings inflation
  - Reform that tightens cap. req's $\rightarrow$ tilt into private ratings
  - Private *ratings* demand $\rightarrow$ private debt *issuance*

- Regulators should take private markets seriously when deciding policy

  **Potential solutions:**
  - Ensure review *before* allowing proposed capital charges
  - Require a public credit rating alongside, or multiple sources
  - Regulator-driven modeling (AI?)

# Thank you!

**dtwennin@iu.edu**

# Appendix

# Fact 3: Include Insurer-Bond Pairs with Time Varying Controls

$$(\text{RBC Risk Factor})_{ibt} = \beta\ (\textbf{Private Rating})_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

| | | | | | |
|---|---|---|---|---|---|
| Private Rating | 0.69*** | −3.16*** | −3.89*** | −2.03*** | −2.24*** |
| | (0.07) | (0.18) | (0.30) | (0.44) | (0.18) |
| Year FE | ✓ | ✓ | | | |
| Issuer FE | | ✓ | | | |
| Issuer × Year FE | | | ✓ | ✓ | |
| Bond Controls | | | | ✓ | |
| Bond FE | | | | | ✓ |
| Maturity FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 434,411 | 426,154 | 364,006 | 189,173 | 385,778 |
| Within $R^2$ | 0.001 | 0.01 | 0.008 | 0.009 | 0.012 |

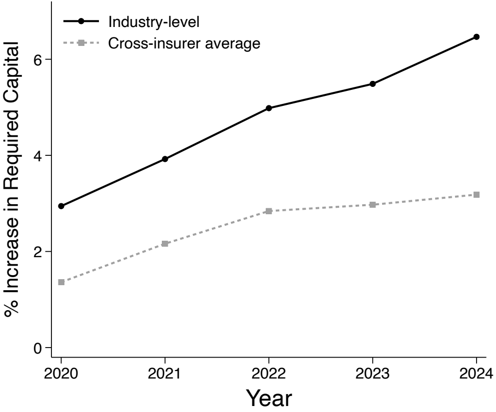
## Fact 3: Full Insurer-Bond Pair Sample

$$\text{(RBC Risk Factor)}_{ibt} = \beta\ \textbf{(Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

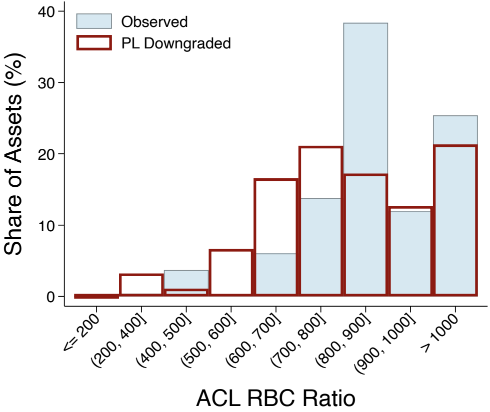
| | | | | | |
|---|---|---|---|---|---|
| Private Rating | 0.73*** | −3.04*** | −3.77*** | −1.97*** | −2.20*** |
| | (0.07) | (0.17) | (0.27) | (0.41) | (0.16) |
| Year FE | ✓ | ✓ | | | |
| Issuer FE | | ✓ | | | |
| Issuer × Year FE | | | ✓ | ✓ | |
| Bond Controls | | | | ✓ | |
| Bond FE | | | | | ✓ |
| Maturity FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 460,133 | 452,060 | 389,137 | 199,495 | 411,535 |
| Within $R^2$ | 0.001 | 0.01 | 0.008 | 0.009 | 0.012 |

▸ Back

# Additional Downgrade Counterfactual Results




(a) Additional Aggregate Req. Capital




(b) RBC Ratios

# RBC Reform Robustness

$$\text{SharePL}_{jt} = \beta \text{ Exposed}_j \times \text{Post}_t + \alpha_t + \delta_j + \varepsilon_{jt}$$

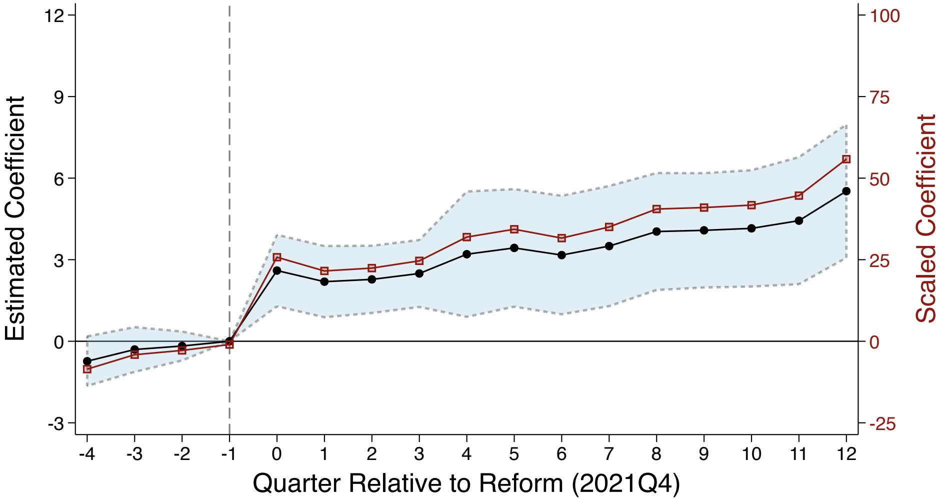
| | Q4 only | PP control | CV weights |
|---|---|---|---|
| Exposure × Post | 5.82*** | 3.73*** | 5.25 |
| | (1.25) | (0.91) | (1.12) |
| Quarter FE | ✓ | ✓ | ✓ |
| Insurer FE | ✓ | ✓ | ✓ |
| Size + ROA Controls | ✓ | ✓ | ✓ |
| Observations | 3,329 | 11,356 | 11,341 |
| Within $R^2$ | 0.04 | 0.38 | 0.03 |

# RBC Reform: Heterogeneity by Financial Constraints

$$\text{SharePL}_{jt} = \beta \text{ Exposed}_j \times \text{Post}_t + \alpha_t + \delta_j + \varepsilon_{jt}$$

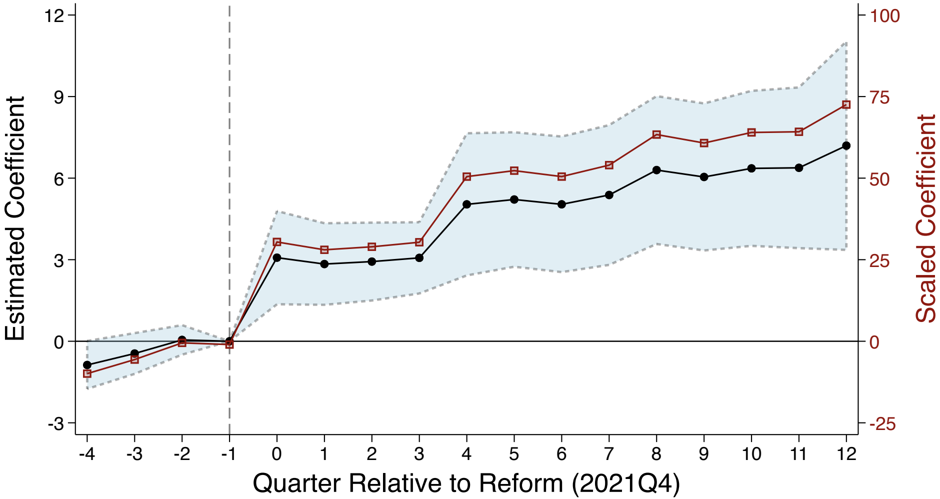
| | | | | |
|---|---|---|---|---|
| Exposure $\times$ Post | 6.78** | 1.77*** | 3.37*** | 0.90*** |
| | (2.88) | (0.63) | (0.99) | (0.33) |
| Constrained? | ✓ | | ✓ | |
| 2.5% Trimmed Sample? | | | ✓ | ✓ |
| Quarter FE | ✓ | ✓ | ✓ | ✓ |
| Insurer FE | ✓ | ✓ | ✓ | ✓ |
| Size + ROA Controls | ✓ | ✓ | ✓ | ✓ |
| Observations | 5,857 | 5,448 | 5,622 | 5,388 |
| Within $R^2$ | 0.02 | 0.02 | 0.02 | 0.03 |

# RBC Reform Robustness: Private Placement Share Control



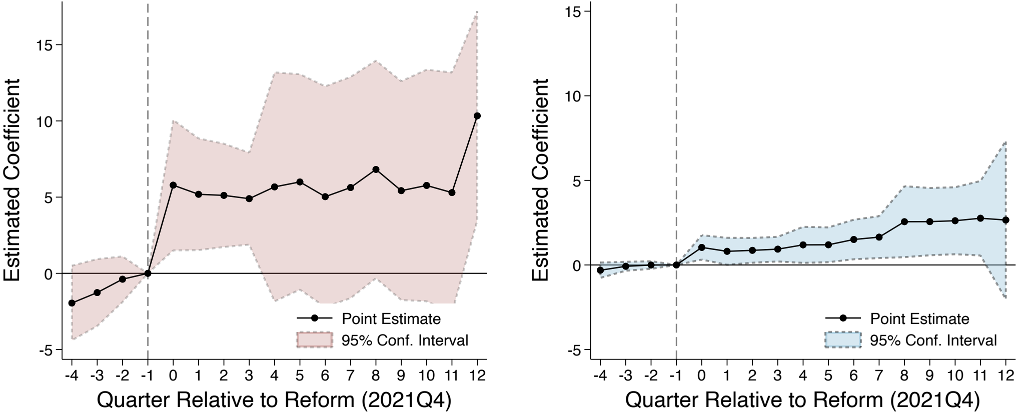

‣ Back

# RBC Reform Robustness: CV-weighted Reform Measure




▸ Back

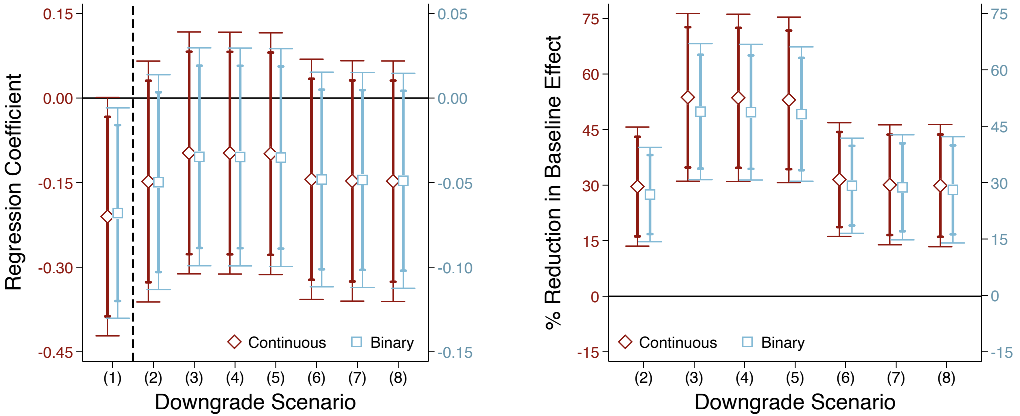
# RBC Reform: Heterogeneity by Financial Constraints




(a) Below-median RCS

(b) Above-median RCS

# Did Private Ratings Attenuate the Effect of the Reform?

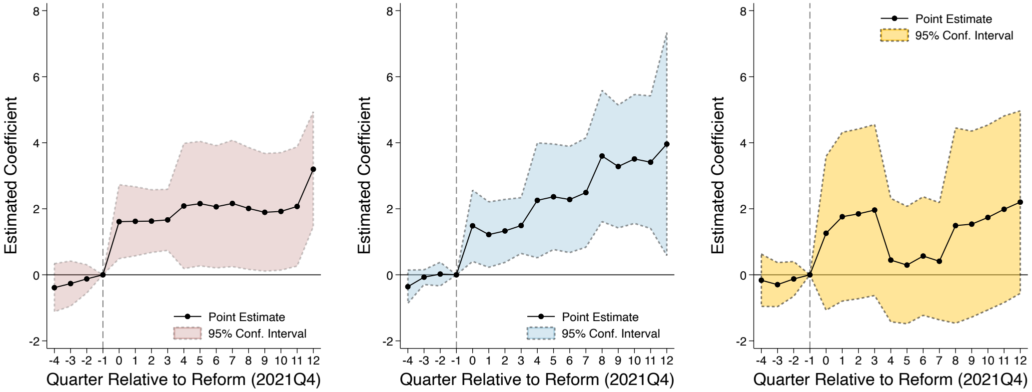



(a) Regression Estimates

(b) Implied Attenuation of PLR Rebalancing

‣ Back

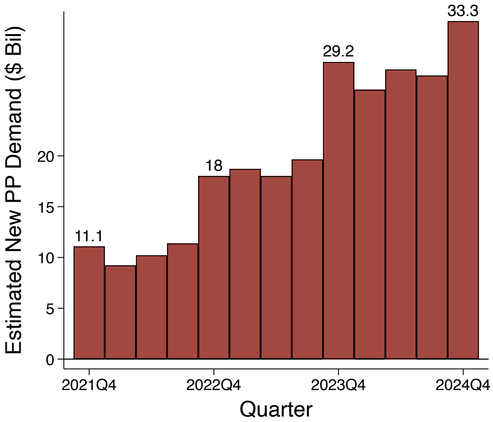
# Did the Reform Create Demand for Ratings or Placement?



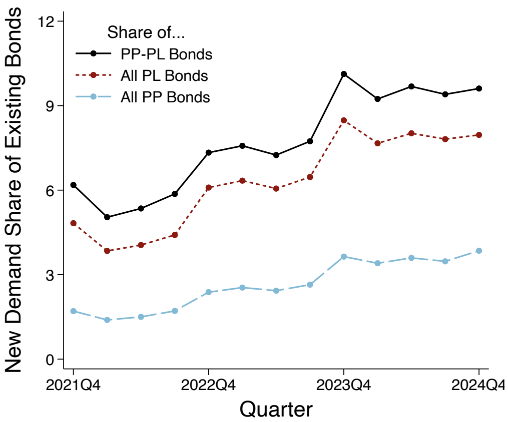

(a) Public bond, private rating (b) Private bond, private rating (c) Private bond, public rating

▸ Back

# Did the Reform Create Demand for Ratings or Placement?




(a) Cumulative New Demand ($Bil)




(b) Cumulative Demand Share of Holdings

Back

# Likelihood of Newly Acquired PP's Receiving PLR

$$\text{AnyPL}_{bt} = \beta\ (\text{Private Placement})_b + \alpha_t + \varepsilon_{bt}$$

| | 0 quarters | $\leq$ 4 quarters | $\leq$ 8 quarters |
|---|---|---|---|
| Privately Placed | 1.78*** | 1.61*** | 1.46*** |
| | (0.40) | (0.38) | (0.39) |
| % Rel. to non-PP | 493% | 400% | 331% |
| Quarter FE | ✓ | ✓ | ✓ |
| Observations | 8,371 | 5,874 | 4,012 |
| BIC | 596.3 | 686.9 | 591.7 |
| Pseudo $R^2$ | 0.09 | 0.08 | 0.05 |

Back